Institutional design and the stability of collective deterrence

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Abstract

Collective deterrence in a coalition of states depends on two key elements: the institutional design of decisionmaking, formalized as a social choice function that aggregates member states levels of resolve, and the strength of the coalitions available means of escalation. This paper studies the interaction of these elements using a multiactor signalling model, with a particular focus on the tradeoff between credible deterrence and the risk of false positives. The model offers a simple formal framework for analyzing how small alliancessuch as the Nordic countries, which we use as an illustrative examplecould, in theory, structure their decision rules to sustain credible deterrence even in the absence of external guarantees.

1 Introduction

In essence, deterrence relies on a push-pull between two fundamental objectives that are, in effect, the strategic analogues of Type I and Type II errors CITE :

1. the resolve to use retaliatory means have to be sufficiently credible, and
2. one should limit the possibility of escalation spirals and accidental use.

The dynamic between these two mechanisms is especially potent for nuclear deterrence, where the consequences are potentially of an existential nature. For a classical outlook on these fundamental issues see CITE and CITE .

NATO is a nuclear alliance, where the deterrence capabilities of individual nations are also shared to allies. However, NATO does not own nuclear weapons as a collective entity, nor does it operate a joint nuclear command structure with shared launch authority. For example, U.S. weapons stationed in Europe remain under exclusive U.S. custody and control, and any nuclear mission requires authorization from the relevant national leaders in addition to political approval through NATO's Nuclear Planning Group (NPG). CITE

Hence, there is currently no coalition of states in the world that jointly own both the deterrence capabilities and the command structure to act upon threats, as the historical prposals for a NATO multinational nuclear force never materialized CITE . However, such coalitions might become more politically viable now that arms-control frameworks, such as the New START treaty CITE , are intensifying the European discussions about burden sharing and the possible future role of France or the UK in European nuclear deterrence.

Currently, of course, the Non-Proliferation Treaty CITE serves as a block to such nuclear alliances, but one could consider other deterrence technologies not under such restrictions. In the advent of such coalition then, the additional problem of *collective* rather than *unilateral* decision-making arises to complicate the above two problems. The problem of *credible resolve* becomes a problem of how the coalition's states aggregate their individual resolves into a joint decision, and the problem of *escalation control* becomes a problem of not allowing this joint decision to be too easily affected by individual potential errors. Hence this joint institutional decision-structure introduces a trade-off between the two fundamental problems. Such aggregations of individual choices is in social choice theory measured by a *social choice function*, see CITE . We can then phrase the motivating question for the present paper as: How does the institutional design, in the form of a social choice function, of a coalition affect the deterrence mechanism and its effectiveness?

To answer this question we develop a simple choice function-dependent signalling model, and characterize the resulting credibility-false positive frontier, and discuss implications for a small alliance: the Nordics.

2 Theoretical background

Our model exists as a combination of two structures: signalling games and social choice functions. We first describe the latter, and then

2.1 Signalling games

Signalling games are dynamic Bayesian games, first studied in relation to learning by Lewis CITE . The players have incomplete information about the initial situation, and the game unfolds sequentially over time, with actions known to the other players CITE .

More precicely, one player – often called the sender – has some private information about the state of the world, an takes an observable action S, called the signal, after which another player – often called the reciever – updates their beliefs about the world and chooses an action. In our case of deterrence, incomplete information is essential; the uncertainties around the opponents resolve lies at the heart of the mechanism making deterrence a viable strategy.

2.2 Social choice functions

3 The model

Our model adopts this basic structure of a signalling game, as presented above, where we assume one of the players to be a coalition of multiple states $\mathcal{C} = \{1, 2, \ldots, n\}$. The other player is some potential attacking adversarial actor $\mathcal{A}$. The adversary $\mathcal{A}$ has a *type*, $T \in \{0, 1\}$, where 1 represents being agressive – actively planning an attack – and 0 represents a benign adversary. The coalition $\mathcal{C}$ holds a common prior $\pi = p(T = 1) \in (0, 1)$, describing their belief that $\mathcal{A}$ has type 1.

If the adversary $\mathcal{A}$ attacks, i.e. $T = 1$, each member i of the coalition $\mathcal{C}$, recieves a binary warning signal $s_i \in \{0, 1\}$, giving a probability

$$p_i = p(s_i = 1 \mid T = 1).$$

If the adversary is of type $T = 0$, then coalition members recieve false alarms with probability

$$q_i = p(s_i = 1 \mid T = 0).$$

We assume that signals are independent across members.

Each coalition member $i \in \mathcal{C}$ then decides on a binary vote $v_i \in \{0, 1\}$, interpreted as their vote towards authorizing retaliation.

In order to focus this paper on the institutional mechanisms, and not on wether the coalition members try to strategize for their own gain, we assume that $v_i = s_i$, which means that we assume the actors to vote truthfully based on their own intelligence assessments.

Ulike a classical signalling game, the coalition does not decide on a signal. The signal is *implied* by the institutional rule that determines the outcome of the voting process based on the individual coalition members' votes.

3.1 Weighted threshold functions

The coalition is assumed to have a publically known decision rule, formalized by a social choice function f. In order to be able to analyze the behaviour of our signalling game with respect to such a function, we need to restrict to a class of viable rules. Fist, we assume a binary outcome, 1 representing coalition retaliation and 0 representing no retaliation. Second, we assume the choice function f to be a weighted threshold rule. This means that each coalition member i is assigned a weight $w_i \in \mathbb{R}$ dependent on the context of the attack, and that there is a threshold $\tau \in \mathbb{R}$, such that the aggregated vote of the coalition is 1 if the weighted sum of the individual members votes exceeds the threshold. This weight scheme is meant to include the possibility that the attacked country in the coalition gets a vote that counts more towards the total, or that frontline countries have a larger say, or that a country has far better ISR capabilities etc.

Formally, then, given a list of votes coalition individual votes $v = (v_1, v_2, \ldots, v_n)$, and a weight scheme $w = (w_1, \ldots, w_n)$, we have

$$f(v; w) = \begin{cases} 1, & \sum_{i=1}^{n} w_i v_i \geqslant \tau \\ 0, & \text{else} \end{cases}$$

Most viable voting rules are formalizable in this framework, like the majority rule, consensus rule, dictatorship, qualified majority, veto players, etc. There are also some decision rules that are not possible to model in this framework, like multi-stage rules, lexicographical rules, non-monotonic rules and randomized rules. However, such rules are very unlikely to be viable for choosing a retaliation action, hence our focus on weighted threshold rules.

3.2 Retaliation probabilities

Assuming the attacker type is $T = 1$, retaliation will occur if the weighted number of coalition members recieving the signal meets the threshold. Because these signals are independent, the probability of retaliation is

$$R(\tau) = p\left(\sum_{i=1}^{n} w_i v_i \geqslant \tau \mid T = 1\right).$$

This function captures the *deterrence credibility* of the coalition $\mathcal{C}$. In other words it is the probability that an attack by the adversary $\mathcal{A}$ triggers a retaliatory response.

Similarly, if the adversary's type is $T = 0$, then retaliation is accidental, with probability

$$F(\tau) = p\left(\sum_{i=1}^{n} w_i v_i \geqslant \tau \mid T = 0\right).$$

This function captures the coalition's *escalation risk*, i.e., the probability that the coalition accidentaly authorizes a response when there had been no attack. This could happen if, for example, another actor than $\mathcal{A}$ was responsible for an attack on the coalition, or that an accident is interpreted as an attack.

Notice that a lowe threshold τ increases the probability of both deterrence and accidental use.

In the simple case that each coalition member i has an equal weight, say $w_i = 1$ for all i, and the threshold τ is a positive integer less than or equal to n, then these probabilities become binomial, with

$$R(\tau) = \sum_{k=\tau}^{n} \binom{n}{k} p^k (1-p)^{n-k},$$

and similarly

$$F(\tau) = \sum_{k=\tau}^{n} \binom{n}{k} q^k (1-q)^{n-k}.$$

The often used majority rules and consensus rules are both of this nature, and these closed formula allows us to study these in slightly more detail.

3.3 Costs and benefits

4 Equilibria and implications

5 The Nordics

6 Conclusion